Semantic Compatibility and Global Concurrence for Boolean Satisfiability Exact Contractions, Certified Section Selectors, and Deterministic Difference-Map Closure

Jan Mikulík

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Abstract

We develop a unified mathematical representation of Boolean satisfiability in which a CNF formula is converted into a finite system of locally admissible states linked by exact compatibility relations. A satisfying assignment is proved to be equivalent to a global section of this compatibility system. This representation separates three logically different operations that are often conflated in SAT algorithms: exact semantic contractions, certified section selectors, and global concurrence dynamics.

Exact contractions preserve the complete solution space and include EXACT1 fusion, alternating even-cycle macrostates, affine elimination over $\mathbb{F}_2$, and finite-field coordinate changes. Certified section selectors need not preserve all solutions; instead they are partial deterministic constructions whose output is accepted only when it is proved to be a global section. Ordered occurrence-balance closure and a trimmed-parity construction are examples of this second type.

For global concurrence we formulate a generalized difference map using deterministic retractions onto two constraint sets. We prove that every fixed point yields an intersection point, and more strongly that convergence is unnecessary for soundness: if any iterate produces a retracted state lying in both constraint sets, that state is already an exact certificate. This principle is instantiated in three formula-derived coordinates: multi-partition EXACT1 concurrence, parity-noise/cardinality concurrence, and an EXACT1–BDD scheduling concurrence. We give explicit formulas and proofs for the local retractions, including a weighted Gaussian retraction onto an affine syndrome manifold and a dynamic-programming retraction onto a capacity BDD.

The resulting framework explains why a small numerical update norm is not a sufficient stopping criterion: a stationary continuous field may still have nonzero compatibility deficiency. The mathematically relevant terminal condition is membership in the global section set itself.

The constructions in this paper are sound partial SAT procedures. They do not constitute a proof that every satisfiable CNF is solved. The remaining completeness problem is stated explicitly: whether a polynomially bounded deterministic semantic compiler and global concurrence map can recover a section for every satisfiable CNF. A proof of that statement would yield a polynomial-time SAT algorithm and therefore imply $P = NP$; no such conclusion is claimed here.

1 Introduction

Let

$$F = C_1 \wedge \cdots \wedge C_m$$

be a CNF formula over Boolean variables

$$X = \{x_1, \ldots, x_n\}.$$

The conventional formulation of SAT asks for a point

$$x \in \{0, 1\}^n$$

that satisfies all clauses simultaneously. This presentation emphasizes the ambient Boolean cube. The present work emphasizes a different object: the family of local satisfying state spaces and the exact consistency relations on their overlaps.

For each local region we retain only admissible local states. If two regions contain a common Boolean variable, the corresponding local states are compatible precisely when they assign the same value to the shared variable. The entire formula then becomes a finite compatibility geometry. The basic equivalence is

$$\boxed{\mathrm{SAT}(F) \cong \Gamma(\mathfrak{K}(F))}$$

where $\mathfrak{K}(F)$ is a canonical compatibility system and $\Gamma(\mathfrak{K}(F))$ denotes its global sections.

This reformulation does not solve SAT by itself. It does, however, isolate the actual global problem: local admissibility is easy; the hard step is to glue locally admissible states into one globally compatible section.

The framework developed below has five layers:

$$\boxed{F \longrightarrow \mathfrak{K}(F) \longrightarrow \widetilde{\mathfrak{K}}(F) \longrightarrow \omega \longrightarrow \text{closure/concurrence} \longrightarrow \sigma \in \Gamma(\mathfrak{K}(F)).}$$

Here $\widetilde{\mathfrak{K}}(F)$ is an exact semantic coordinate system when one is available, ω is a formula-derived continuous or discrete orientation, and the final closure step is accepted only after an exact section criterion is reached.

The principal contributions are as follows.

(i) We define a canonical local-compatibility system for every CNF and prove a bijection between satisfying assignments and global sections.

(ii) We separate *exact semantic contractions*, which preserve the entire solution set, from *certified section selectors*, which may select only a subset but whose successful output is rigorously satisfying.

(iii) We formulate a generalized difference map for arbitrary deterministic retractions and prove its fixed-point and finite-iterate intersection certificates.

(iv) We derive exact local evaluators for OR and EXACT1 factors and explain why the universal state-space representation need not be explicitly exponential.

(v) We formalize three global-concurrence coordinates used by the current implementation: multi-partition categorical concurrence, parity-noise/cardinality concurrence, and BDD scheduling concurrence.

(vi) We state precisely what remains unproved: universal deterministic completeness for all satisfiable CNF formulas.

The implementation motivating the present formulation is deterministic and does not use benchmark names, local-search flips, residual-guided repair, stochastic restart portfolios, or an external SAT solver in the active constructions discussed here. SAT is emitted only after one independent exact verification of the original CNF. This implementation contract guarantees sound emitted SAT answers, but it is not a completeness theorem.

2 Canonical local compatibility systems

2.1 Local admissible states

For a clause C_a, let $V_a \subseteq X$ be the set of variables occurring in it.

Definition 2.1 (Local satisfying state space)**.** The local satisfying state space of C_a is

$$S_a = \{s : V_a \to \{0,1\} \mid C_a(s) = 1\}.$$

For two regions a, b with nonempty overlap, define

$$K_{ab} : S_a \times S_b \to \{0,1\}$$

by

$$K_{ab}(s_a, s_b) = 1 \iff s_a|_{V_a \cap V_b} = s_b|_{V_a \cap V_b}.$$

Thus K_{ab} contains no heuristic information: it is exact equality on the shared Boolean coordinates.

Definition 2.2 (Finite compatibility system)**.** A finite compatibility system is a tuple

$$\mathfrak{K} = (\mathcal{R}, \{S_a\}_{a\in\mathcal{R}}, E, \{K_{ab}\}_{(a,b)\in E}),$$

where $\mathcal{R}$ is a finite set of regions, S_a is the local state space of region a, E is a set of interacting region pairs, and K_{ab} is the exact compatibility relation on each interaction.

Definition 2.3 (Global section)**.** A global section is a choice

$$\sigma = (s_a)_{a\in\mathcal{R}}, \qquad s_a \in S_a,$$

such that

$$K_{ab}(s_a, s_b) = 1 \qquad \forall (a,b) \in E.$$

The set of all global sections is denoted $\Gamma(\mathfrak{K})$.

2.2 Canonical clause–variable cover

For every Boolean variable x_i introduce a two-state variable region

$$B_i = \{0,1\}.$$

For every clause C_a retain its satisfying local state space S_a. Connect the clause region a to the variable region B_i whenever $x_i \in V_a$. Compatibility requires that the value of x_i inside the clause state equal the value selected by B_i.

The resulting bipartite system is denoted $\mathfrak{K}(F)$.

Theorem 2.4 (Universal compatibility representation)**.** *For every CNF formula F there is a canonical bijection*

$$\Phi : \mathrm{SAT}(F) \longrightarrow \Gamma(\mathfrak{K}(F)).$$

Consequently,

$$F \text{ is satisfiable} \iff \Gamma(\mathfrak{K}(F)) \neq \varnothing.$$

Proof. Let $x \in \mathrm{SAT}(F)$. On variable region B_i select state x_i. On clause region a select the restriction

$$s_a = x|_{V_a}.$$

Because x satisfies F, each $s_a \in S_a$. Every clause state agrees with every incident variable state by construction, so the resulting family is a global section.

Conversely, let $\sigma \in \Gamma(\mathfrak{K}(F))$. The variable regions define a unique Boolean assignment $x \in \{0,1\}^n$. Compatibility forces every local clause state to equal $x|_{V_a}$. Since that local state belongs to S_a, clause C_a is true. Hence every clause is true and $x \in \mathrm{SAT}(F)$.

The two constructions are inverse. Therefore Φ is a bijection. □

Remark 2.5. Theorem 2.4 is a representation theorem, not an algorithmic completeness result. It relocates the difficulty from searching $\{0,1\}^n$ to recovering a global compatible section.

2.3 Compatibility deficiency

A useful continuous diagnostic is obtained by placing a probability distribution p_a on each S_a. For a directed interaction set $E^{\to}$ define

$$\mathcal{D}(p) = \frac{1}{2} \sum_{(a,b)\in E^{\to}} \sum_{u\in S_a} \sum_{v\in S_b} p_a(u)\,[1 - K_{ab}(u,v)]\,p_b(v).$$

Proposition 2.6 (Nonnegativity and discrete exactness)**.** *For all local distributions p, $\mathcal{D}(p) \geq 0$. If every p_a is a point mass at s_a, then*

$$\mathcal{D}(p) = 0 \iff (s_a)_a \in \Gamma(\mathfrak{K}).$$

Proof. Every summand is nonnegative, proving $\mathcal{D}(p) \geq 0$. For point masses, each directed edge contributes either 0 or 1 according to whether the chosen pair is compatible. Therefore the sum is zero exactly when every interacting pair is compatible, which is precisely the definition of a global section. □

This proposition explains a central empirical lesson: a small update norm

$$\|H_{t+1} - H_t\|$$

does not imply $\mathcal{D} = 0$. Numerical stationarity and logical concurrence are different notions.

3 Exact semantic contractions

The canonical clause–variable cover is universal but redundant. Structured CNF encodings frequently contain groups of clauses that are exactly equivalent to a smaller semantic relation.

Definition 3.1 (Exact semantic contraction)**.** Let $\mathfrak{K}$ and $\mathfrak{K}'$ be compatibility systems over the same decoded Boolean variables. An exact semantic contraction is a deterministic representation map

$$Q : \mathfrak{K} \to \mathfrak{K}'$$

for which there is an exact reconstruction map inducing a bijection

$$\Gamma(\mathfrak{K}) \cong \Gamma(\mathfrak{K}')$$

with the same decoded Boolean assignments.

Proposition 3.2 (Composition)**.** *A finite composition of exact semantic contractions is an exact semantic contraction.*

Proof. Bijections compose. If each stage preserves and reconstructs global sections exactly, their composition does as well. □

3.1 EXACT1 fusion

Lemma 3.3 (ALO plus complete AMO equals EXACT1)**.** *For Boolean variables $x_1, \ldots, x_k$,*

$$(x_1 \vee \cdots \vee x_k) \wedge \bigwedge_{1\leq i<j\leq k} (\neg x_i \vee \neg x_j)$$

is equivalent to

$$\sum_{i=1}^{k} x_i = 1.$$

Proof. The positive clause requires at least one true variable. Every binary negative clause forbids a pair of simultaneous true variables, and the complete pair family therefore permits at most one true variable. Hence exactly one is true. The converse is immediate. $\square$

The entire clause family can therefore be replaced by a categorical region with states

$$S = \{e_1, \dots, e_k\},$$

where e_j selects exactly variable x_j.

3.2 Alternating even-cycle contraction

Let $B_0, \dots, B_{\ell-1}$ be pairwise disjoint variable bundles, with ℓ even, and let

$$G_j = B_{j-1} \sqcup B_j \qquad (j \bmod \ell)$$

be EXACT1 regions. Put

$$y_j = \sum_{x \in B_j} x.$$

Since G_j is EXACT1,

$$y_{j-1} + y_j = 1.$$

Proposition 3.4 (Even-cycle macrostates)**.** *If ℓ is even, the bundle-level variables have exactly two patterns,*

$$(1, 0, 1, 0, \dots) \quad \textit{and} \quad (0, 1, 0, 1, \dots).$$

Whenever $y_j = 1$, exactly one Boolean variable in B_j is true.

Proof. From $y_{j-1} + y_j = 1$ we obtain $y_j = 1 - y_{j-1}$, so the sequence alternates. Even cycle length makes both choices of y_0 consistent after one circuit. EXACT1 also forces $y_j \in \{0, 1\}$ and, if $y_j = 1$, exactly one member of bundle B_j is selected. $\square$

Thus a large cycle may be represented by two macro-orientations plus local categorical choices.

3.3 Affine elimination over $\mathbb{F}_2$

Suppose an exact semantic subsystem is

$$Ax = b \qquad \text{over } \mathbb{F}_2.$$

Gaussian elimination may express pivot variables as affine functions of free variables z.

Proposition 3.5 (Affine coordinate preservation)**.** *Assume exact elimination gives*

$$x_i = c_i \oplus \bigoplus_{j \in J_i} z_j$$

for every eliminated variable. Then substitution and back-substitution induce a bijection between solutions in the free coordinates and solutions of the original affine subsystem.

Proof. Elementary row operations over a field preserve the solution set. Once pivots are fixed, every assignment of free variables determines all pivot variables uniquely by back-substitution. Conversely, every solution of the original system restricts to a free-coordinate assignment that reconstructs the same solution. $\square$

The important distinction is that affine elimination changes coordinates but does not choose a solution.

3.4 Finite-field APN coordinates

A particularly strong coordinate system arises for odd m over $\mathrm{GF}(2^m)$.

Theorem 3.6 (Gold cubic)**.** *Let m be odd. The map*

$$f(x) = x^3$$

on $\mathrm{GF}(2^m)$ is a permutation and is almost perfect nonlinear: for every $a \neq 0$ and every b, the equation

$$f(x+a) + f(x) = b$$

has at most two solutions.

Proof. Because m is odd,

$$2^m \equiv 2 \pmod 3, \qquad 2^m - 1 \equiv 1 \pmod 3,$$

so

$$\gcd(3, 2^m - 1) = 1.$$

Hence exponentiation by 3 permutes the nonzero multiplicative group and fixes 0.

For the differential equation, characteristic two gives

$$(x+a)^3 + x^3 = ax^2 + a^2x + a^3.$$

For $a \neq 0$, set $y = x/a$ and divide by a^3:

$$y^2 + y = \frac{b}{a^3} + 1.$$

The $\mathbb{F}_2$-linear map $L(y) = y^2 + y$ has kernel $\{0, 1\}$, so every nonempty fiber has exactly two elements. Therefore the differential equation has at most two solutions. □

Corollary 3.7. *Invertible linear changes of input or output coordinates preserve permutation and APN properties.*

When a formula-derived finite-field chart certifies complete pairwise ALLDIFFERENT relations, affine two-plane no-collapse relations, and full-rank anchors, Theorem 3.6 supplies a closed-form section candidate. The current implementation reconstructs the Boolean variables by exact $\mathbb{F}_2$ back-substitution and verifies the original CNF once. This is a certified special chart, not a general completeness theorem.

4 Certified section selectors

Exact contractions preserve all global sections. A second operation is useful and logically different.

Definition 4.1 (Certified section selector)**.** A certified section selector is a partial deterministic map

$$S : \mathfrak{K} \rightharpoonup \Gamma(\mathfrak{K})$$

such that whenever $S(\mathfrak{K})$ is defined, its output is rigorously a global section. The selector may be undefined on satisfiable instances and need not preserve all sections.

This distinction is essential. A rule may impose additional orientation constraints that are not logical consequences of the original formula, provided it accepts them only when they lead to an independently certified satisfying section.

4.1 Exact unit closure

Let a partial Boolean assignment be

$$\omega : D_\omega \to \{0,1\}, \qquad D_\omega \subseteq X.$$

Repeatedly apply ordinary unit propagation to the original CNF until no new variable is forced.

Theorem 4.2 (Total conflict-free unit closure)**.** *If unit propagation from ω terminates without contradiction and assigns every Boolean variable, then the resulting total assignment satisfies F.*

Proof. Let x be the total closure. Suppose some clause were false. Since x is total, the clause would contain zero true literals and zero unassigned literals. During unit propagation this is exactly a conflict, contradicting the hypothesis. Hence every clause is true. □

4.2 Ordered occurrence-balance orientation

For an ordered CNF presentation, traverse clauses and literals in their written order. Let

$$N_i^+(t), \qquad N_i^-(t)$$

be the positive and negative occurrence counts seen before position t.

At a non-final literal position, if the current literal has occurred at least as often as its opposite polarity, orient that literal to false. Thus a positive literal proposes $x_i = 0$ and a negative literal proposes $x_i = 1$. If contradictory orientations occur, reject the chart. If the resulting partial orientation has total conflict-free unit closure, Theorem 4.2 certifies the result.

Proposition 4.3 (Trace selector soundness)**.** *Whenever ordered occurrence-balance orientation is consistent and its unit closure is total and conflict-free, the produced assignment satisfies the original CNF.*

Proof. Immediate from Theorem 4.2. □

Warning 4.4 (Presentation dependence)*.* The trace depends on clause and literal order. Two syntactically permuted CNFs representing the same Boolean function may produce different trace orientations. Therefore this is a presentation-derived certified selector, not an exact semantic contraction.

4.3 Three-of-four parity completion is not an equivalence

This point is important for mathematical correctness.

For three variables x, y, z, an XOR relation

$$x \oplus y \oplus z = r$$

accepts four assignments and rejects the four assignments of parity $1 - r$. Each rejected assignment can be forbidden by one ternary clause.

Lemma 4.5 (Complete ternary parity CNF)**.** *A set of four ternary clauses on the same three variables, whose falsifying assignments are exactly the four assignments of parity $1 - r$, is equivalent to*

$$x \oplus y \oplus z = r.$$

Proof. Each clause forbids exactly one Boolean assignment. The four clauses jointly forbid precisely all assignments of parity $1 - r$, leaving exactly the four assignments of parity r. □

Warning 4.6. Three of those four clauses do *not* imply the XOR equation. One opposite-parity assignment remains legal. Completing a three-clause group to an XOR equation is therefore a restriction of the solution space, not an exact semantic contraction.

The restriction is nevertheless usable as a certified selector when its remaining low-dimensional coordinates are checked exactly against the original formula.

Theorem 4.7 (Rank-$n-1$ trimmed-parity selector)**.** *Suppose a formula on n variables yields $n-1$ consistent ternary parity hypotheses, each arising from either a complete four-clause parity group or a three-of-four group with a common parity orientation. Suppose the resulting affine system has rank $n-1$. Then its solution set contains exactly two assignments $x^{(0)}, x^{(1)}$.*

Define for each original clause C the allowed free-coordinate set

$$A_C = \{q \in \{0,1\} : C(x^{(q)}) = 1\}.$$

If

$$A = \bigcap_{C \in F} A_C \neq \varnothing,$$

then every $q \in A$ yields a satisfying assignment of the original CNF.

Proof. Rank $n-1$ leaves one free affine coordinate, hence exactly two affine assignments. By definition, $q \in A$ belongs to every A_C, so every original clause is true under $x^{(q)}$. Therefore $x^{(q)} \in \mathrm{SAT}(F)$. □

This is not candidate ranking. The free coordinate is a two-point symbolic domain, and the algorithm computes the exact intersection of the subsets allowed by every original clause.

5 Exact local partition responses

The canonical state space of a width-k clause contains $2^k - 1$ satisfying local assignments. The mathematical representation may therefore be exponentially large if materialized naively. The relevant local partition responses, however, can often be evaluated without enumerating those states.

5.1 Universal compatibility response

For a directed compatibility edge $a \to b$, local field $h_{a\to b} : S_a \to \mathbb{R}$, and inverse temperature $\beta > 0$, define

$$\mathcal{T}^{(\beta)}_{a\to b}[h](v) = \frac{1}{\beta} \log \sum_{u \in S_a} K_{ab}(u, v) e^{\beta h_{a\to b}(u)}.$$

This is the same abstract operation for every finite compatibility relation. Semantic charts differ only in how the sum is represented and evaluated.

As $\beta \to \infty$,

$$\frac{1}{\beta} \log \sum_j e^{\beta z_j} \to \max_j z_j,$$

so the partition response approaches a compatibility max-projection.

5.2 OR factor

Consider

$$C_a = \ell_1 \vee \cdots \vee \ell_k$$

and let $c_{ai} \in \{+1, -1\}$ denote literal polarity. If $L_{i \to a}$ is a Boolean log-field, define

$$p_{i \to a}^{\text{viol}} = \sigma(-c_{ai} L_{i \to a}), \qquad \sigma(t) = \frac{1}{1 + e^{-t}}.$$

Conditioned on the target variable i, the clause is violated precisely when every other literal is false. Hence

$$q_{a \setminus i} = \prod_{j \neq i} p_{j \to a}^{\text{viol}}.$$

The exact log-response is

$$U_{a \to i} = c_{ai} \big[-\log(1 - q_{a \setminus i}) \big].$$

Proposition 5.1 (Polynomial OR evaluation)**.** *All outgoing OR responses for a clause of width k can be evaluated in polynomial time in k without enumerating $2^k - 1$ local satisfying states.*

Proof. The responses depend only on products of violation probabilities excluding one factor. These can be obtained from a total product with guarded division, or from prefix–suffix products, in $O(k)$ or $O(k^2)$ arithmetic depending on implementation. Both are polynomial in k. □

5.3 EXACT1 factor

For

$$x_1 + \cdots + x_k = 1,$$

let e^{L_j} be the incoming relative weight of state $x_j = 1$. Conditioned on $x_i = 1$, every other variable is zero, so after common normalization

$$Z_i(1) = 1.$$

Conditioned on $x_i = 0$, exactly one $j \neq i$ is true, so

$$Z_i(0) = \sum_{j \neq i} e^{L_j}.$$

Therefore

$$U_{a \to i} = \log Z_i(1) - \log Z_i(0) = -\log \sum_{j \neq i} e^{L_j}.$$

Again no enumeration of all Boolean assignments is needed.

6 Global concurrence by a generalized difference map

The central new closure mechanism is expressed as an intersection problem between two constraint sets.

Let V be a finite-dimensional real vector space and let

$$A, B \subseteq V$$

be constraint sets. We do not require convexity. Let

$$R_A : V \to A, \qquad R_B : V \to B$$

be deterministic retractions or projection-like maps. Metric nearest-point optimality is not required for the soundness statements below; only membership of their outputs in A and B is required.

For $\beta \neq 0$, define

$$f_A(x) = \left(1 - \frac{1}{\beta}\right) R_A(x) + \frac{x}{\beta},$$
$$f_B(x) = \left(1 + \frac{1}{\beta}\right) R_B(x) - \frac{x}{\beta},$$

and the difference map

$$\mathrm{DM}_\beta(x) = x + \beta \left[R_A(f_B(x)) - R_B(f_A(x))\right].$$

This is the standard difference-map geometry specialized to the parameter convention used here.

Theorem 6.1 (Fixed-point intersection certificate)**.** *If x^* is a fixed point of DM_β, then*

$$y^* = R_A(f_B(x^*)) = R_B(f_A(x^*))$$

and therefore

$$y^* \in A \cap B.$$

Proof. The fixed-point equation gives

$$0 = \mathrm{DM}_\beta(x^*) - x^* = \beta \left[R_A(f_B(x^*)) - R_B(f_A(x^*))\right].$$

Since $\beta \neq 0$,

$$R_A(f_B(x^*)) = R_B(f_A(x^*)).$$

The common value belongs to A because it is an output of R_A, and to B because it is an output of R_B. □

More important for SAT soundness is the following finite-iterate statement.

Theorem 6.2 (Finite-iterate intersection certificate)**.** *For any iterate x, define*

$$a = R_A(f_B(x)).$$

If an exact internal test establishes $a \in B$, then

$$a \in A \cap B$$

regardless of whether x is a fixed point and regardless of whether the difference-map sequence converges.

Proof. By construction a is an output of R_A, hence $a \in A$. The additional exact test gives $a \in B$. Thus $a \in A \cap B$. □

Remark 6.3. Theorem 6.2 is the key soundness principle used by the current concurrence charts. Convergence is an efficiency and completeness question, not a prerequisite for correctness of a detected intersection.

The implementation uses the fixed constant

$$\beta = \frac{11}{20} = 0.55$$

for the new concurrence charts. No theorem in this paper claims that this value is universally optimal or universally convergent.

7 Multi-partition EXACT1 concurrence

Consider categorical regions $G_1, \dots, G_r$. Region G_a has states indexed by the variables it may select, and every valid local state is one-hot.

Introduce one real slot for each pair (a, s) consisting of a region and one of its local states. Let

$$\nu(a, s) \in \{1, \dots, n\}$$

be the underlying Boolean variable represented by that slot.

7.1 Local constraint set

Define

$$A = \{z : \text{each region block is one-hot}\}.$$

For a real block z_a, define R_A by selecting an index of maximal value and setting that coordinate to 1 and every other coordinate in the block to 0.

Proposition 7.1 (One-hot projection)**.** *With deterministic tie-breaking, blockwise argmax is a Euclidean nearest-point projection onto the one-hot set.*

Proof. For a block $z \in \mathbb{R}^k$ and one-hot vector e_j,

$$\|z - e_j\|_2^2 = \|z\|_2^2 + 1 - 2z_j.$$

Minimizing the distance is therefore equivalent to maximizing z_j. □

7.2 Concurrence subspace

All slots representing the same Boolean variable must agree. Define

$$B = \{z : z_{a,s} = q_{\nu(a,s)} \text{ for some } q \in \mathbb{R}^n\}.$$

If variable i has c_i copies, the orthogonal projection is

$$(R_B z)_{a,s} = \frac{1}{c_i} \sum_{(b,t):\nu(b,t)=i} z_{b,t}, \qquad i = \nu(a, s).$$

Proposition 7.2 (Concurrence projection)**.** *The copywise mean map is the orthogonal projection onto B.*

Proof. For each variable i, the coordinates belonging to its copy class form an independent Euclidean block. The least-squares constant approximating a finite list is its arithmetic mean. □

Theorem 7.3 (Exact-cover intersection)**.** *Assume every Boolean variable is represented by at least one slot. A point $a \in A$ belongs to B if and only if, for every Boolean variable, either all of its copies are selected or none are selected. Such points are in bijection with Boolean assignments satisfying every categorical EXACT1 region.*

Proof. Because $a \in A$, every slot is Boolean and every region has exactly one selected slot. Membership in B requires all copies of each underlying variable to have the same value, hence either all copies are 1 or all are 0. Define x_i to be that common value. Each region has exactly one selected variable, so x satisfies every EXACT1 region.

Conversely, a Boolean assignment satisfying all EXACT1 regions defines a one-hot slot vector. Every copy of a Boolean variable receives its common Boolean value, so the vector belongs to B. □

The implementation checks this intersection criterion exactly by counting selected copies. No original CNF residual is inspected during concurrence. Only after a global section is found is the original CNF independently verified.

8 Parity-noise/cardinality concurrence

We now describe a second coordinate system in which exact parity chains collapse to a noisy linear observation model.

8.1 Exact component collapse

Suppose complete parity clause groups have first been converted, using Lemma 4.5 and its higher-width analogue, into exact equations over $\mathbb{F}_2$. Consider one connected component of these equations. Assume internal auxiliary variables occur an even number of times in the component, while one distinguished variable r_j occurs once. XORing every equation in the component cancels the even-occurrence auxiliaries and yields

$$a_j^T s \oplus r_j = y_j.$$

Collecting m such components gives

$$As \oplus r = y, \qquad s \in \mathbb{F}_2^n, \quad r \in \mathbb{F}_2^m.$$

Proposition 8.1 (GF(2) component cancellation)**.** *If an auxiliary variable occurs an even number of times among equations XORed together, it disappears from the resulting equation. Variables with odd total incidence remain.*

Proof. Over $\mathbb{F}_2$, adding a variable to itself gives zero. Thus its coefficient in the XOR-sum equals its incidence count modulo two. □

The strict implementation additionally recognizes a unary counter geometry from the formula itself and recovers a unique tolerated corruption weight t. The semantic target becomes

$$\boxed{As \oplus r = y, \qquad \|r\|_0 \le t.}$$

8.2 Affine retraction

Write

$$M = [A \mid I_m]$$

and

$$u = (s, r) \in \mathbb{F}_2^{n+m}.$$

Then the exact parity set is

$$A_{\mathrm{par}} = \{u \in \{0,1\}^{n+m} : Mu = y\}.$$

Given $z \in \mathbb{R}^{n+m}$, threshold to

$$b_i = \mathbf{1}[z_i \ge 1/2].$$

Its syndrome is

$$\eta = y \oplus Mb.$$

Order the columns of M from least confident to most confident according to

$$\kappa_i = |1 - 2z_i|.$$

Run Gaussian elimination on the columns in that order until a full row-rank basis has been collected. Solve for a correction vector c satisfying

$$Mc = \eta$$

and set

$$R_{\mathrm{par}}(z) = b \oplus c.$$

Theorem 8.2 (Weighted Gaussian retraction)**.** *The construction above always returns a point in A_{par}, provided the selected correction columns span $\mathbb{F}_2^m$. In the present augmented system $[A \mid I_m]$, such a full-rank set always exists.*

Proof. By construction $Mc = \eta = y \oplus Mb$. Therefore

$$M(b \oplus c) = Mb \oplus Mc = Mb \oplus (y \oplus Mb) = y.$$

Hence $b \oplus c \in A_{\text{par}}$. The identity block contains all standard basis columns, so the full column family spans $\mathbb{F}_2^m$. □

The confidence ordering chooses which valid affine correction is produced. The theorem claims retraction into the affine set, not Euclidean nearest-point optimality.

8.3 Cardinality retraction

Define

$$B_t = \{(s, r) \in \{0,1\}^{n+m} : \|r\|_0 \le t\}.$$

Threshold all coordinates. If at most t corruption coordinates are 1, retain them. Otherwise retain the t corruption coordinates with largest z_i and set the remaining corruption coordinates to zero.

Proposition 8.3 (At-most-t projection)**.** *The preceding map is a Euclidean nearest-point projection of z onto B_t, with deterministic tie-breaking.*

Proof. Without the cardinality bound, coordinatewise thresholding is the nearest Boolean vector. If more than t corruption coordinates prefer 1, exactly t may remain 1. The squared-distance advantage of assigning coordinate i the value 1 rather than 0 is

$$z_i^2 - (1 - z_i)^2 = 2z_i - 1,$$

which increases monotonically with z_i. Therefore the t largest preferred coordinates must be retained. □

8.4 Concurrence certificate

Apply the difference map with

$$R_A = R_{\text{par}}, \qquad R_B = R_t.$$

At an inspection step let

$$a = R_{\text{par}}(f_B(x)).$$

By Theorem 8.2, a satisfies the exact parity equations. If

$$\|a_r\|_0 \le t,$$

then $a \in B_t$ as well. Theorem 6.2 therefore gives an exact intersection point.

The selected s and r coordinates are then inserted into the original CNF as a partial assignment and exact unit closure is performed.

Theorem 8.4 (Parity-noise selector soundness)**.** *If the parity-noise difference map produces a with $Ma = y$ and $\|a_r\|_0 \le t$, and exact unit propagation from the corresponding partial assignment is total and conflict-free, then the reconstructed Boolean assignment satisfies the original CNF.*

Proof. The difference-map state supplies a consistent partial assignment. The conclusion follows from Theorem 4.2. □

Every operation in one iteration is polynomial: thresholding and cardinality projection are polynomial, and Gaussian elimination is polynomial in $n + m$ and m. The implementation uses a linear iteration cap in the coordinate dimension, subject to a fixed minimum.

9 BDD scheduling concurrence

A third coordinate system arises when a formula decomposes completely into one-hot assignment groups, machine-capacity BDDs, monotone negative pruning relations, assignment-pair exclusions, and fixed units.

The crucial requirement is a *complete clause grammar certificate*: the chart is activated only if every original clause belongs to one of the recognized exact classes.

9.1 Job EXACT1 set

Let $J_1, \dots, J_q$ be disjoint groups of assignment variables, one group per job. Define

$$A_{\text{job}} = \left\{ z \in \{0,1\}^N : \sum_{i \in J_j} z_i = 1 \quad \forall j \right\}.$$

The retraction R_J independently chooses the maximum coordinate in each job group. By the one-hot projection proposition, this is a Euclidean nearest-point projection.

9.2 Capacity BDDs

Consider a rooted ordered binary decision diagram with false terminal 0, true terminal 1, and an internal node v labelled by assignment variable $i(v)$ with children $L(v), R(v)$ corresponding to decisions $0, 1$.

For a real vector z, define terminal costs

$$d(0) = +\infty, \qquad d(1) = 0.$$

For internal node v define

$$d(v) = \min \left\{ z_{i(v)}^2 + d(L(v)), (1 - z_{i(v)})^2 + d(R(v)) \right\}.$$

Backtracking from the root chooses the branch attaining the minimum.

Theorem 9.1 (BDD dynamic-programming projection)**.** *Assume the BDD is an ordered acyclic decision diagram and each tested variable occurs at most once along a root-to-terminal path. The dynamic program above returns a Boolean assignment following an accepting root-to-true path with minimum squared distance contribution from the constrained decision variables.*

Proof. Proceed by induction from the terminals upward. The terminal values are correct: the false sink is infeasible and the true sink requires no further cost. At an internal node every accepting completion must choose either the 0 branch or the 1 branch. The two displayed terms are exactly the immediate squared-distance costs plus the optimal continuation costs supplied inductively by the corresponding child. Taking the smaller therefore gives the optimal accepting completion below the node. Backtracking reconstructs it. □

Coordinates not constrained on the selected reduced-BDD path may be thresholded independently.

9.3 Monotone pruning closure

The recognized additional constraints have the forms

$$\neg x_i \vee \neg x_j$$

and

$$\neg x_i \vee \neg a_1 \vee \cdots \vee \neg a_k,$$

where the a_j are active BDD auxiliary states. They are monotone in the assignment variables: changing an assignment bit from 1 to 0 cannot violate a capacity constraint of the form

$$\sum_i w_i x_i \leq C.$$

Proposition 9.2 (Deletion monotonicity)**.** *If x satisfies a nonnegative weighted capacity inequality and $0 \leq y \leq x$ coordinatewise, then y also satisfies it.*

Proof. Since $w_i \geq 0$,

$$\sum_i w_i y_i \leq \sum_i w_i x_i \leq C.$$

□

The machine retraction therefore proceeds as follows: first retract independently onto accepting BDD paths; next eliminate any simultaneously selected forbidden assignment pair; then repeatedly recompute active BDD auxiliary states and delete an assignment choice whenever an all-negative pruning clause is violated. Each nontrivial closure step changes at least one assignment bit from 1 to 0, so the deletion process terminates after at most N such deletions.

Proposition 9.3 (Machine-set retraction)**.** *Under the recognized capacity-BDD grammar and monotone negative pruning grammar, the machine closure returns a Boolean assignment satisfying every machine BDD, every assignment-pair exclusion, and every pruning relation.*

Proof. The initial BDD dynamic program yields accepted machine assignments. Pair repair only deletes 1 bits, so capacity feasibility is preserved by deletion monotonicity and every processed pair becomes legal. The subsequent pruning closure also only deletes assignment bits. Active BDD paths are recomputed after deletions until no pruning rule fires. Hence the final state satisfies all pruning relations and remains capacity-feasible. Deleting bits cannot create a new pairwise 11 violation. □

Let B_{mach} denote this machine/pruning constraint set.

9.4 Job–machine difference map

The global concurrence problem is

$$A_{\text{job}} \cap B_{\text{mach}}.$$

The implementation initializes the continuous assignment coordinates from a formula-derived factor field through a sigmoid; no intermediate Boolean candidate is used for orientation. It then applies the difference map with

$$R_A = R_J, \qquad R_B = R_M.$$

At inspection steps, the locally one-hot state

$$a = R_J(f_B(x))$$

is tested directly against all machine BDDs, assignment-pair relations, and pruning relations.

Theorem 9.4 (BDD concurrence certificate)**.** *Assume the semantic compiler partitions every original CNF clause into: exact job-EXACT1 clauses, canonical capacity-BDD clauses, recognized monotone pruning clauses, assignment-pair exclusions, or fixed units. If a difference-map iterate produces $a \in A_{\text{job}}$ that passes every machine/pruning constraint, then the Boolean assignment obtained from a, the active BDD paths, and the fixed units satisfies the complete original CNF.*

Proof. Membership in A_{job} satisfies every exact job-EXACT1 relation. Passing the machine test means each machine BDD reaches its true terminal and all monotone pruning and assignment-pair exclusions hold. Active BDD paths determine the auxiliary-node variables consistently, and fixed units are assigned their prescribed values. By hypothesis these clause classes form a partition of the original clause set. Hence every original clause is satisfied. $\square$

The implementation nevertheless performs one independent final CNF verification after reconstruction.

10 A unified section-recovery architecture

The constructions above can be expressed with one common language.

10.1 Layer I: universal representation

Every CNF has the canonical object

$$F \mapsto \mathfrak{K}(F)$$

from Theorem 2.4.

10.2 Layer II: exact coordinate changes

Whenever exact structure is recognized, one may apply

$$\mathfrak{K}(F) \xrightarrow{Q} \widetilde{\mathfrak{K}}(F)$$

with

$$\Gamma(\mathfrak{K}(F)) \cong \Gamma(\widetilde{\mathfrak{K}}(F)).$$

Examples include EXACT1 fusion, even-cycle macrostates, exact affine elimination, and certified finite-field coordinates.

10.3 Layer III: orientation

A formula-derived orientation may be discrete or continuous:

$$\omega \in \Omega(\widetilde{\mathfrak{K}}).$$

It is not itself a satisfying assignment. Examples include an occurrence-balance partial assignment, an all-zero parity-noise state, or a continuous factor field for the BDD concurrence chart.

10.4 Layer IV: closure or concurrence

One then applies an exact closure operator or an intersection dynamics:

$$\omega \longmapsto \mathrm{Cl}(\omega) \quad \text{or} \quad x_{t+1} = \mathrm{DM}_\beta(x_t).$$

The terminal condition is not a small numerical update. The terminal condition is an exact section certificate.

10.5 Layer V: Boolean reconstruction

Once a semantic section σ is certified, decode it to

$$x \in \{0,1\}^n$$

and verify the untouched original CNF.

This yields the conceptual chain

$$\boxed{F \to \mathfrak{K}(F) \to \widetilde{\mathfrak{K}}(F) \to \omega_F \to \text{Closure/Concurrence} \to \sigma_F \to x_F.}$$

11 Master soundness statements

Theorem 11.1 (Soundness of exact-chart intersection). *Suppose a semantic chart has an exact decoder*

$$D : A \cap B \to \mathrm{SAT}(F).$$

If a deterministic difference-map iteration produces an intersection certificate $a \in A \cap B$, then $D(a)$ is a satisfying assignment.

Proof. Immediate from the definition of the decoder. □

Theorem 11.2 (Soundness of certified-selector reconstruction). *Suppose a chart produces a partial or restricted semantic state and then performs either:*

(a) exact symbolic intersection with every original clause, or

(b) total conflict-free unit closure of the original CNF.

Then every emitted assignment satisfies the original CNF.

Proof. Case (a) is true by direct clause intersection. Case (b) is Theorem 4.2. □

Corollary 11.3 (Independent-verifier soundness). *If an implementation emits SAT only after directly evaluating every original clause on the final assignment and observing zero violated clauses, then every emitted SAT answer is sound independently of the behavior of the preceding continuous dynamics.*

This corollary is elementary but operationally important: it separates *soundness* from *completeness*. A continuous method may fail to find a section; it must never label a non-section SAT.

12 Complexity of the certified components

The framework contains several operations whose polynomial complexity is explicit.

- Canonical CNF parsing and verification are linear in total literal incidence.
- EXACT1 recognition by a complete AMO clique is polynomial in the local group width.
- Gaussian elimination over $\mathbb{F}_2$ is polynomial.
- The rank-$n-1$ trimmed-parity selector constructs two affine assignments and checks all original clauses, hence is polynomial.
- One multi-partition concurrence iteration is linear in the number of categorical slots plus copy incidences.

- One parity-noise retraction uses polynomial Gaussian elimination and an at-most-t selection.
- One BDD dynamic-programming pass is linear in the number of BDD nodes; the monotone pruning closure performs at most linearly many deletions, each with polynomial recomputation.

The current implementation also imposes polynomial iteration caps on the new difference-map charts. Therefore these chart procedures are polynomially bounded partial algorithms.

What is *not* proved is that one of them succeeds on every satisfiable CNF, or that the generic compatibility transport always reaches a section.

13 Preliminary validation

The present paper is theory-first. The following table records a preliminary regression set used to test the mathematical distinctions developed above. Interrupted long-running instances are deliberately excluded. A zero in the final column means that the independently verified final assignment violated no original clause.

Instance	Variables	Clauses	Active semantic coordinate	Final U
uf250-0100	250	1065	generic clause–variable cover	0
170058440	320	1120	certified ordered trace	0
ac-sgen1-sat-180-100	180	432	multi-partition EXACT1 concurrence	0
battleship-14-26-sat	364	2562	disjoint AMO/OR witness chart	0
frb65-12-2	780	24731	categorical EXACT1 cover	0
mdp-28-11-sat	799	4438	parity-noise/cardinality concurrence	0
pcmax-scheduling-m15	2351	13561	EXACT1–BDD concurrence	0
rbsat-v1150c84314	1150	84314	categorical EXACT1 cover	0
two-trees-511v	511	2039	rank-$n-1$ trimmed-parity selector	0
x9-12035	600	5406	alternating even-cycle chart	0
apn-sbox5-cut3	21240	86081	finite-field APN chart	0

Four of these instances directly motivated the present concurrence formulation. Before the new global closure layer, their final residual counts were respectively

$$1, \quad 5, \quad 35, \quad 41$$

for the multi-partition, parity-noise, BDD scheduling, and trimmed-parity cases. The important observation was not merely that residual clauses remained, but that the failures had different geometric causes:

(i) a multi-partition field could become numerically stationary while retaining incompatible copies of the same Boolean variable;

(ii) an affine field could retain large symmetry manifolds with exact zero coordinates;

(iii) a heterogeneous native factor flow could be severely under-scaled relative to the true semantic dimension;

(iv) a locally valid affine contraction could still lack a canonical global orientation.

These failures motivated the replacement of "small update norm" by "exact intersection/global section" as the acceptance criterion.

The table is not evidence of universal completeness. Its purpose is regression and falsification: each new semantic law must preserve previously solved structurally different instances while eliminating identified failure mechanisms.

14 The unresolved global completeness problem

The exact representation theorem and the sound partial charts do not settle SAT complexity. The missing statement can now be formulated precisely.

Conjecture 14.1 (Universal deterministic section recovery)**.** *There exists a deterministic polynomial-time semantic compiler and a polynomially bounded sequence of exact contractions, formula-derived orientations, local compatibility evaluations, and global concurrence/closure maps such that, for every satisfiable CNF formula F, the procedure returns*

$$\sigma \in \Gamma(\mathfrak{K}(F)).$$

A formulation closer to the difference-map picture asks for polynomially computable constraint representations A_F, B_F, deterministic retractions R_{A_F}, R_{B_F}, an initialization $x_0(F)$, and a polynomial bound $T(|F|)$ such that

$$A_F \cap B_F \neq \varnothing \Longrightarrow \exists t \leq T(|F|): \quad R_{A_F}(f_{B_F}(x_t)) \in A_F \cap B_F.$$

No proof of this implication is presently known.

Theorem 14.2 (Conditional complexity consequence)**.** *If Conjecture 14.1 holds with a total polynomial bound known from the input size, then*

$$P = NP.$$

Proof. Given any CNF formula F, run the deterministic procedure for its guaranteed polynomial bound and verify the returned assignment directly. If F is satisfiable, the conjecture guarantees that a satisfying assignment is recovered within the bound. Therefore failure to recover one within the bound certifies that F is unsatisfiable. SAT would thus be decidable in deterministic polynomial time. Since SAT is NP-complete, $P = NP$. □

Remark 14.3. The known implementation is not a proof of Conjecture 14.1. In particular, empirical success on many structured instances cannot replace a worst-case convergence theorem.

15 Why stationary fields are insufficient

The original motivation for global concurrence can be expressed as a simple logical gap.

Suppose a continuous transport reaches

$$\|H_{t+1} - H_t\| \ll 1.$$

This only says that H_t is approximately stationary for the chosen numerical map. It does not imply

$$\mathcal{D}(H_t) = 0,$$

and it does not imply that independent local argmax states glue to a global section.

The desired universal statement would instead have the form

$$\boxed{\Gamma(\mathfrak{K}) \neq \varnothing, \quad \mathcal{D}(H) > 0 \Longrightarrow \text{the deterministic dynamics cannot terminate at } H.}$$

A proof of such a theorem for a polynomially bounded dynamics would be far stronger than a convergence-by-update-norm criterion.

The new concurrence charts avoid this logical error locally: they terminate only when an exact intersection predicate is true. What remains open is whether a universal version of the same principle can be constructed for arbitrary satisfiable CNF formulas.

16 Status ledger

For clarity, the mathematical status of the principal claims is summarized below.

Statement	Status
Every CNF admits the canonical clause–variable compatibility system	proved
SAT assignments are in bijection with global sections	proved
EXACT1 fusion is an exact semantic contraction	proved
Alternating even EXACT1 cycles admit exact macrostates	proved
Exact Gaussian elimination preserves affine solution spaces	proved
Gold x^3 is a permutation and APN over $\mathrm{GF}(2^m)$ for odd m	proved
Total conflict-free unit closure is sound	proved
Four ternary parity clauses of one parity orientation are equivalent to one XOR relation	proved
Three-of-four parity completion is an exact equivalence	false; explicitly not claimed
Rank-$n-1$ three/four-clause parity completion plus exact symbolic closure is a sound selector	proved
Difference-map fixed points yield intersection points	proved
A finite iterate whose retracted state belongs to both constraint sets is an exact certificate	proved
Block argmax and copy averaging implement the stated multi-partition projections	proved
Weighted Gaussian correction retracts onto the affine syndrome manifold	proved
Top-t corruption selection projects onto the at-most-t Boolean set	proved
BDD dynamic programming gives an optimal accepted-path decision under the stated grammar	proved
Monotone deletion preserves nonnegative capacity feasibility	proved
Every satisfiable CNF reaches a certified intersection in polynomially many steps	open
SAT is in P	not claimed
$P = NP$	not claimed

17 Limitations and research directions

Several limitations are structural rather than cosmetic.

No universal convergence theorem. The difference map supplies a clean intersection certificate, but the present work does not prove that it reaches such a certificate on every satisfiable instance.

Finite semantic grammar. The current semantic compiler recognizes several exact or certified coordinate systems. The universal canonical compatibility representation remains available when no stronger chart is recognized, but the generic transport is not proved complete.

The parameter β. The value $\beta = 11/20$ is fixed across the new concurrence charts in the implementation, but no universal optimality theorem is claimed.

Selectors versus contractions. Some successful constructions deliberately restrict the solution space. The trimmed-parity construction is the clearest example. Such methods must be described as certified selectors, not as equivalence-preserving rewrites.

Floating-point transport. Some generic and legacy semantic flows use specified floating-point arithmetic. Their numerical trajectories are implementation objects, not exact mathematical proofs. Soundness of emitted SAT is protected by exact final Boolean verification.

BDD grammar specialization. The BDD chart is activated only when the entire original CNF can be parsed into the certified clause grammar. This is a strong structural condition, not a theorem that every scheduling CNF admits the same parser.

Worst-case iteration counts. Polynomial iteration caps make the partial algorithms bounded, but observed large iteration counts motivate a separate study of contraction rates, invariant geometry, and improved deterministic retractions.

The most important theoretical direction is to identify a compatibility functional or topological invariant whose positive value rules out non-solution stationary points on satisfiable systems.

18 Conclusion

The central exact identity of this work is

$$\boxed{\mathrm{SAT}(F) \cong \Gamma(\mathfrak{K}(F)).}$$

It provides a universal mathematical object for Boolean satisfiability: locally admissible states connected by exact compatibility relations.

Within this object we distinguished three operations that must not be conflated:

$$\boxed{\text{exact contraction} \neq \text{certified section selection} \neq \text{global concurrence dynamics.}}$$

Exact contractions preserve all solutions. Certified selectors may restrict the solution set but accept only rigorously satisfying outputs. Difference-map concurrence searches for an intersection of two exact semantic constraint sets and is sound whenever an exact intersection is detected, even without a convergence proof.

The multi-partition, parity-noise, and BDD constructions demonstrate that apparently different SAT encodings can be expressed by the same abstract pattern:

$$\boxed{\text{local admissibility} \cap \text{global concurrence} = \text{certified section.}}$$

The trimmed-parity construction shows a complementary route in which a strong affine hypothesis reduces the problem to an exact low-dimensional symbolic closure.

The resulting framework explains a key failure mode of purely continuous SAT dynamics: numerical stationarity does not imply logical compatibility. The appropriate terminal object is a global section itself.

What remains open is the decisive question: whether every satisfiable CNF admits a polynomially recoverable canonical orientation and a polynomially bounded deterministic closure to a global section. Proving that statement would prove $P = NP$. The present paper does not claim such a proof; it isolates the required theorem and develops a set of exact constructions with which that theorem can be tested, refined, or falsified.

References

[1] S. A. Cook, "The Complexity of Theorem-Proving Procedures," *Proceedings of the Third Annual ACM Symposium on Theory of Computing*, pp. 151–158, 1971.

[2] R. M. Karp, "Reducibility Among Combinatorial Problems," in *Complexity of Computer Computations*, R. E. Miller and J. W. Thatcher, eds., Plenum Press, pp. 85–103, 1972.

[3] V. Elser, I. Rankenburg, and P. Thibault, "Searching with Iterated Maps," *Proceedings of the National Academy of Sciences*, vol. 104, no. 2, pp. 418–423, 2007.

[4] R. E. Bryant, "Graph-Based Algorithms for Boolean Function Manipulation," *IEEE Transactions on Computers*, vol. C-35, no. 8, pp. 677–691, 1986.

[5] K. Nyberg, "Differentially Uniform Mappings for Cryptography," in *Advances in Cryptology—EUROCRYPT '93*, Lecture Notes in Computer Science, vol. 765, pp. 55–64, Springer, 1994.